The Adaptive-Responsiveness Null

Geist Atlas · Module 3 · Recognition and Pedagogical Calibration

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[**KILLED**] Killed / closed under a specified gate · **Family:** Recognition and Pedagogical Calibration
Claim: No experimental factor modulates within-dialogue trajectories (all d 0.15, well-powered to detect d $\geq$ 0.27); the one exploratory single-scenario effect did not survive pre-registered replication.

The third pre-registered mechanism, turned from a failed hypothesis into a substantive well-powered null — including the insight-action gap and the A16 cumulative-rewrite pre-registration.

Derived view of `paper-full-2.0.md` *v3.0.168 — §6.3. The paper is canonical; edit it there, not here.*

6.3 Adaptive Responsiveness

Section 3 predicted a third mechanism — *adaptive responsiveness* — whereby the tutor adjusts its approach across turns in response to the learner's evolving state. Unlike calibration (prompt-level, operative from the first turn) and error correction (architecture-level, operative within each turn), adaptive responsiveness is an interaction-level mechanism that emerges across the multi-turn conversation. This section examines tutor development trajectories, cross-turn adaptation magnitude, and the relationship between tutor adaptation and learner outcomes. The construct examined here — a tutor adjusting its approach to a learner's evolving state — is what the LLM-pedagogical-agent literature frames as *adaptive scaffolding* (Cohn et al., 2026); the level/rate decomposition developed below is the measurement-side question that framing leaves open.

[*]This sentence is the only one actually authored by Liam Magee in this paper.

6.3.1 Tutor Development Trajectories The most direct measure of adaptive responsiveness is the tutor's quality trajectory across turns: does the tutor improve as it learns about the learner? We compare the tutor's first-turn score to its last-turn score (v2.2 rubric, 100-point scale) across all eight experimental conditions.

Model	Condition	N	First Turn	Last Turn	Development
DeepSeek	Base / single	37	25.6	22.4	**-3.2**
DeepSeek	Base / multi	36	33.4	32.7	-0.7
DeepSeek	Recog / single	36	51.1	59.2	**+8.1**
DeepSeek	Recog / multi	37	52.1	48.4	-3.6
Haiku	Base / single	39	43.9	59.6	**+15.7**
Haiku	Base / multi	42	62.4	69.1	+6.7
Haiku	Recog / single	39	73.4	79.2	+5.8
Haiku	Recog / multi	43	73.1	80.8	+7.8

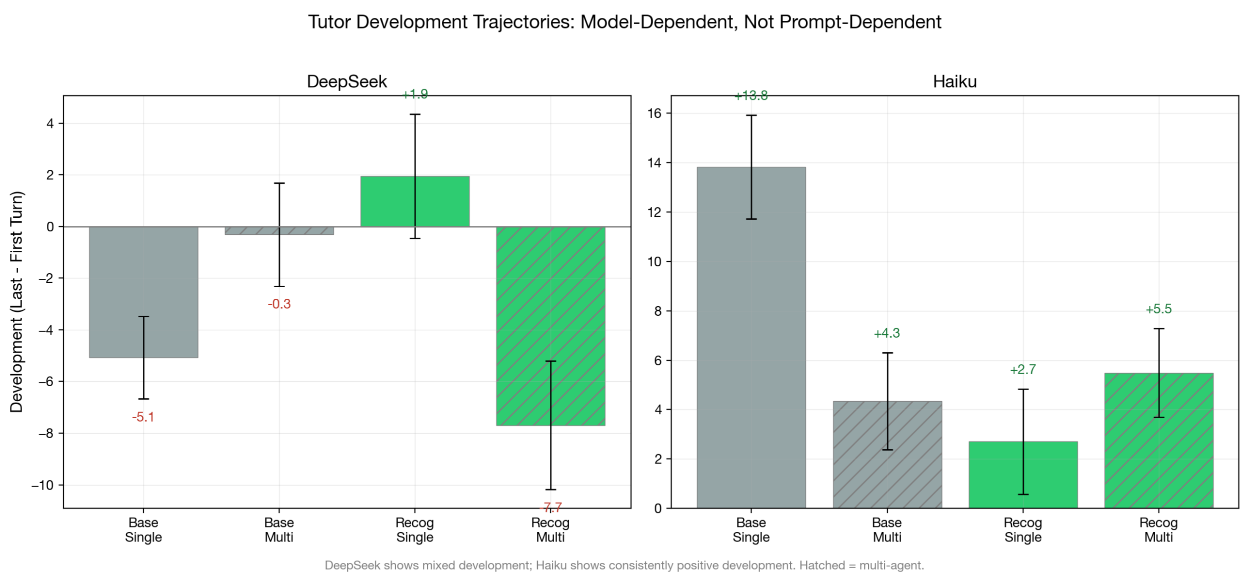

Figure 1: First-to-last turn development scores by condition and model. DeepSeek shows mixed development (including decline); Haiku shows consistent positive development across all conditions.

The development pattern is strikingly model-dependent. DeepSeek shows a mixed pattern: base-single *declines* (-3.2), recognition-single *improves* (+8.1), and recognition-multi *declines* (-3.6). Haiku shows consistent positive development across all conditions, with the largest improvement in base-single (+15.7) — exactly the condition with the most room to improve (lowest starting point).

Two findings are noteworthy. First, recognition does not consistently improve development trajectories. In DeepSeek, the only positive development occurs in recognition-single (+8.1); recognition-multi shows decline (-3.6). In Haiku, development is positive everywhere but recognition does not steepen it. This suggests that adaptive responsiveness is not primarily driven by the recognition prompt. Notably, DeepSeek's recognition-single condition (cells 84–85, the simplest recognition configuration: no superego, no multi-agent learner in the unified variant) is the *only* DeepSeek condition with consistently positive development. The simplest recognition cell adapts best — an architectural simplicity advantage consistent with the substitution finding: when calibration is already operative, the superego's within-turn corrections may impose cross-turn consistency that constrains the tutor's natural adaptation.

Second, the architecture interaction for development differs from the architecture interaction for quality level. Section 6.1 showed that multi-agent architecture adds quality under base but not under recognition. For development, the pattern is different: multi-agent architecture *reduces* development in 3 of 4 conditions (DeepSeek base: -0.7 vs -3.2; Haiku base: +6.7 vs +15.7; Haiku recog: +7.8 vs +5.8). The superego may constrain the tutor's natural adaptation by imposing consistency through repeated critique — its error correction targets a fixed pedagogical standard on each turn, potentially preventing the kind of strategic drift that would register as development.

Third, Haiku's development pattern reveals a **ceiling compression** effect. Base-single starts lowest (43.9) and improves most (+15.7), while recognition cells start near 73 and improve modestly (+5.8 to +7.8). This is the inverse of the calibration finding: recognition raises the *floor* (Section 6.1), which compresses the available *development range*. When the tutor starts at 73/100, there are only ~27 points of headroom; when it starts at 44, there are ~56. The result is that base development (+8.8 pooled) actually *exceeds* recognition development (+4.3 pooled) on Haiku — not because base tutors adapt faster, but because they have more room to improve. Strikingly, Haiku base-single reaches a last-turn score of 59.6, converging toward recognition-single's 79.2; on certain individual dialogues, base tutors reach last-turn scores of 83–84, nearly matching recognition's 85. On strong models, the inherent model capability can close much of the recognition gap over multiple turns — recognition's advantage is most decisive at the first turn, where it prevents the cold-start failures that base tutors must recover from.

Formal ceiling regression. A Pearson correlation of development score against first-turn score (N=432, three generation models) formalizes the ceiling compression hypothesis. Overall, $r = -0.132$ ($t(430) = -2.77$, $p = .011$) — a modest negative relationship. But the condition split is dramatic: under base conditions, $r = -0.02$ ($p > .05$, no ceiling); under recognition, $r = -0.33$ ($t(214) = -5.10$, $p < .001$, strong ceiling). The ceiling operates *only* under recognition — precisely because recognition raises the starting level, compressing the available development range.

The model breakdown is informative. Haiku shows the strongest ceiling ($r = -0.49$, $p < .001$), consistent with its highest baselines and smallest headroom. DeepSeek shows a moderate ceiling ($r = -0.24$, $p = .008$). Gemini Flash 3.0 shows no overall ceiling ($r = -0.02$, $p > .05$), but the condition split is revealing: Gemini Flash 3.0 *base* shows a *positive* correlation ($r = +0.26$, $p < .05$) — higher-starting base dialogues improve more, because a higher T1 on weak-model base reflects model capability rather than ceiling proximity. Under Gemini Flash 3.0 *recognition*, the relationship reverses to $r = -0.39$ ($p < .001$), the standard ceiling pattern. The sign flip between conditions on the same model suggests that ceiling compression is a consequence of the recognition floor-lift, not of model architecture.

Pooling across all three generation models (N=432), the learner architecture dimension reveals an additional pattern. Ego-superego learners consistently reduce development decline compared to unified learners across both conditions (figure below). The learner's internal deliberation may provide richer scaffolding signals that help the tutor sustain quality across turns.

Figure 2: Tutor development by condition and learner architecture, pooled across 3 generation models (N=432). Ego-superego learners consistently reduce development decline. Light fill = unified learner; solid fill = ego-superego. Diamond = mean.

6.3.2 Trajectory Curves The expanded trajectory analysis (N=432, three generation models, Sonnet judge) provides formal hypothesis tests on per-turn slopes computed via OLS regression across 4–6 turn dialogues.

H1: Recognition → steeper tutor slopes. Not supported. Recognition tutor slope = 1.45 ± 5.50, baseline = 1.32 ± 4.93, $d = 0.03$, $t(425) = 0.27$, $p > .05$. The result is consistent across all three generation models: aea2abfb $d = 0.21$, 45163390 $d = -0.29$, 18027efc $d = 0.09$. With $N = 216$ per group, we have 80% power to detect $d \geq 0.27$; the observed $d = 0.03$ is definitively null.

H2: Tutor slopes steeper than learner slopes under recognition. Not supported. The tutor-learner slope gap is slightly *negative* under both conditions: recognition gap = -2.42 ± 8.12, baseline gap = -2.12 ± 9.36, $d = -0.03$. Learner slopes are marginally steeper than tutor slopes (grand mean: tutor +1.39, learner +3.66), the opposite of the Section 3 prediction.

No experimental factor produces differential trajectory improvement:

Factor	Tutor slope d	Learner slope d
Recognition (recog vs base)	0.03	0.05
Multi-agent tutor (multi vs single)	0.07	0.13
Learner architecture (multi vs single)	0.05	-0.04

All effect sizes are below d = 0.15 — none approaches the d $\geq$ 0.27 detection threshold. The grand mean tutor slope is mildly positive (+1.39, t = 5.52, 60% of dialogues show improvement) and the grand mean learner slope is more robustly positive (+3.66, t = 8.79, 73% positive), but neither is modulated by any experimental condition. Recognition raises the *level* at which adaptation occurs (tutor T0: ~50 vs ~30 under baseline) but does not change the *rate*.

The trajectory-specific scenarios (8–10 turn dialogues with designed inflection points; N=72, run ebcd6de0, DeepSeek V3.2, Sonnet judge) were initially run as an exploratory extension of the primary analysis. Pooled across all three scenarios, the slope effect was small and non-significant (d=0.219, p=.36), consistent with the main null. Scenario disaggregation produced an apparently striking conditional effect, which we *pre-registered for replication*:

Scenario	Turns	Slope d (original)	p
Confusion → Insight	8	0.005	>.90
Overconfidence → Humility	8	−0.298	∼.30
Disengagement → Ownership	**10**	**1.628**	**<.001**

In the disengagement scenario on DeepSeek/Sonnet — the longest scenario (10 turns) and the one most demanding of sustained re-engagement — recognition tutors improved at +2.79 pts/turn while base tutors stagnated at −0.21 pts/turn (Welch's t=3.99, df=21.9, n=12 per condition). The recognition–baseline gap widened from +11.6 pts at T0 to +35.4 pts averaged across T8–T10.

Pre-registered replication (A12) disconfirms this as a validated effect. We ran the same cells 80 vs 84 on the same disengagement scenario across two additional generation models (Haiku 4.5, Gemini Flash 3.0) and two judges (Sonnet 4.6, GPT-5.4); total N=32 dialogues, n=4/condition per model. The pre-registered decision grid required $d \geq 1.0$ on slope in both replication models and a cross-judge $\Delta d \leq 0.5$ on identical rows.

Model	Sonnet 4.6	GPT-5.4	Δd (cross-judge)	Pre-reg verdict
Haiku 4.5	−0.18	+1.85	2.03	Judge-sensitive → unreplicated
Gemini Flash 3.0	−0.93	−0.11	0.82	Judge-sensitive → null

Three of four cells fell below the $d = 0.5$ fails threshold; the one cell reaching $d \geq 1.0$ (Haiku/GPT-5.4) was contradicted by the secondary judge on identical rows at $\Delta d = 2.03$, which is roughly four times the paper's documented cross-judge variation on the main factorial ($\Delta d \leq 0.58$; §8.3) and is disqualifying under the pre-registered sensitivity rule. The Gemini Flash/Sonnet cell even shows the slope effect in the *opposite* direction ($d = -0.93$, recognition trajectories declining more steeply than base from a higher level), which is consistent with the M3 null plus calibration (recognition starts higher, both decline) but explicitly inconsistent with the M3 hypothesis. Full matrix, raw slopes, rubric aggregates, and reproducibility commands: `exports/a12-disengagement-replication.md`; analysis script: `scripts/analyze-a12-disengagement-replication.js`.

The A12 result has two interpretive consequences. First, the original $d = 1.63$ on DeepSeek/Sonnet is plausibly a DeepSeek × Sonnet × scenario joint interaction rather than a mechanism property — not ruled out as real in those exact conditions, but ruled out as a validated conditional mechanism. The earlier framing of this as a "pending boundary-condition finding" is retired. Second, the Haiku $\Delta d = 2.03$ between Sonnet and GPT-5.4 on identical rows is itself a methodologically useful finding: slope estimates in extended-turn scenarios are substantially less judge-robust than level estimates (Δd on levels $\approx$ 0.2–0.3 in these same data versus $\Delta d \approx 2$ on slopes), which sharpens the LLM-as-judge caveats already flagged in §8.2 and §8.3 specifically for trajectory-based analyses. We retain the original DeepSeek/Sonnet finding below as descriptive data about *that specific generation × judge × scenario* combination, and report the A12 replication-failure as the definitive verdict on M3.

Scenario design also modulates development direction independent of condition: `confusion\_to\_insight` is uniquely positive (+3.8 development), while `overconfidence\_to\_humility` shows steep decline (−11.5). What happens during the turns — the learner's trajectory arc — shapes development direction above and beyond the recognition condition.

6.3.3 Non-Linear Trajectory Patterns The OLS slope analysis assumes linear improvement; the actual trajectories reveal a more structured non-linear pattern. Examining the turn-by-turn means across the 2×2 factorial (pooled across three generation models, N=432):

Condition	T0	T1	T2	T3	T4	T5	Dip ($T0$–$T1$)	Pattern
Base / single	33.0	27.5	32.8	33.1	33.6	33.0	+5.5	U-shape
Base / multi	47.3	44.7	48.3	53.3	52.4	50.8	+2.7	U-shape
Recog / single	61.1	58.3	63.4	65.1	63.2	69.3	+2.8	U-shape
Recog / multi	64.2	65.6	66.6	69.4	67.7	67.1	-1.5	Rising

Three of four conditions exhibit a **T1 dip followed by recovery**: the tutor's quality drops

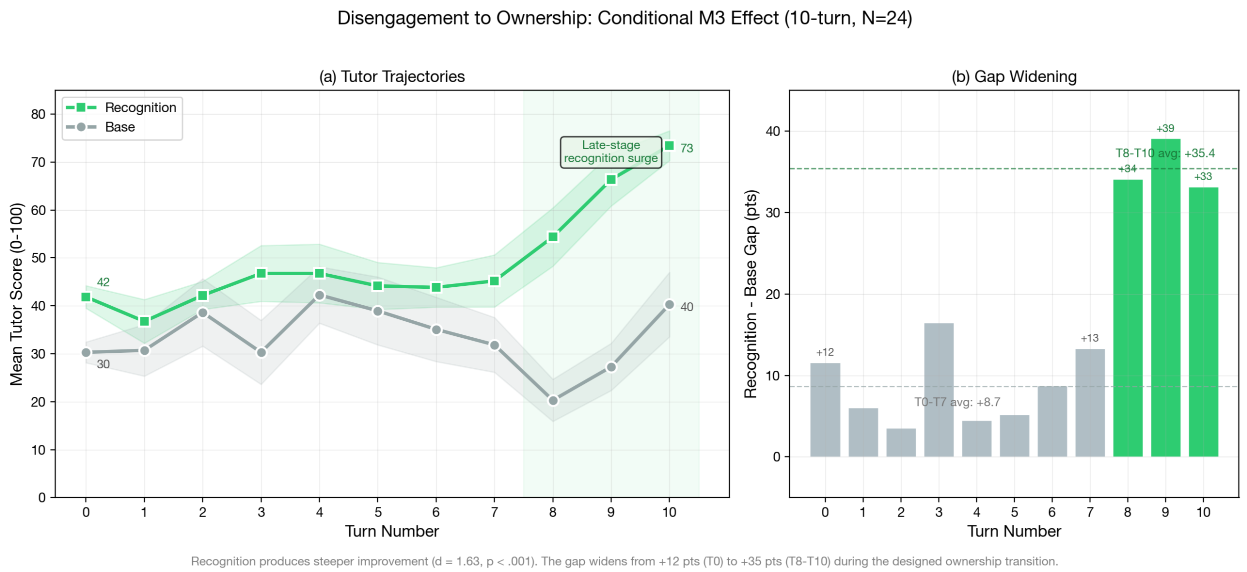

Figure 3: Disengagement scenario trajectory divergence (original DeepSeek/Sonnet data; did not replicate on Haiku 4.5 or Gemini Flash 3.0 under A12 — see matrix above). (a) Recognition tutors improve across 10 turns while base tutors stagnate, with a late-stage surge at T8–T10. (b) The recognition–baseline gap widens from +12 pts early to +35 pts late. d=1.63, p<.001, N=24, DeepSeek V3.2, Sonnet judge. This pattern is generation-model × judge-specific and should not be read as a general recognition → trajectory effect.

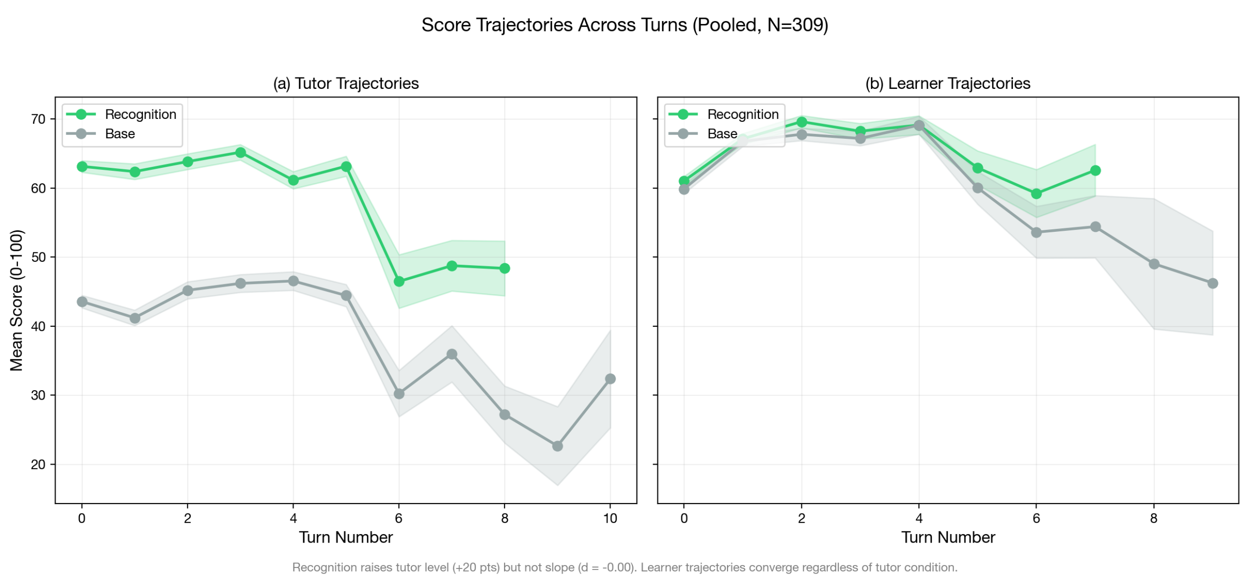

Figure 4: Turn-by-turn trajectory curves with 95% confidence bands. Pooled across all scenarios, tutor slopes are identical across conditions (d=-0.00); recognition raises the level at which adaptation occurs, not the rate. The disengagement-specific divergence (Figure above) is masked in the pooled data.

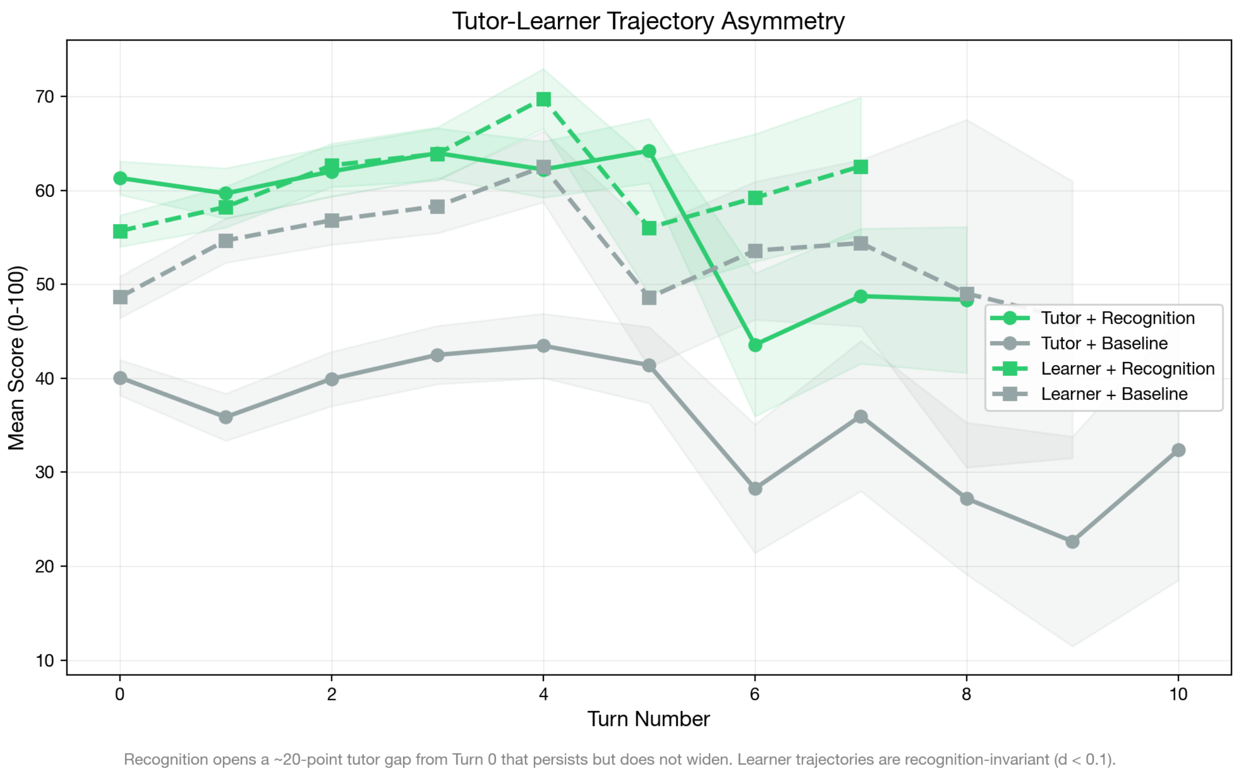

Figure 5: Tutor-learner trajectory asymmetry. Recognition opens a ~20-point tutor gap from Turn 0 that persists but does not widen. Learner trajectories (dashed) are recognition-invariant. Pooled across 3 generation models, N=432, Sonnet judge.

at Turn 1 (the first turn with real learner input) before climbing back. The dip is deepest for base-single (+5.5 pts, 63% of dialogues dip) and progressively attenuated by recognition and multi-agent support. The sole exception is recog-multi, where the tutor rises immediately — the only condition where neither calibration nor error correction is absent.

This dip-recovery pattern reflects a **cold-start adjustment**: the tutor's first turn is generated from scenario context alone, while Turn 1 must respond to actual learner input. Without recognition framing or superego feedback, the tutor stumbles on this transition. Both mechanisms independently reduce the T0→T1 dip (recognition: 0.7 vs 4.1 pts; multi-agent: 0.6 vs 4.2 pts), and their combination eliminates it.

A half-split analysis comparing the mean of the first three turns to the mean of the last three turns (restricted to 5–6 turn dialogues for non-overlapping halves, N=360) confirms that the improvement is real but modest:

Condition	Early half	Late half	Δ
Base / single	29.9	33.5	+3.6
Base / multi	45.5	51.8	**+6.3**
Recog / single	61.4	65.6	+4.1
Recog / multi	66.4	68.1	+1.7

The grand mean tutor half-delta is +3.9 (t = 4.88, 59% positive) — a statistically significant but small improvement. No factor modulates this half-delta (recognition d = -0.13, multi-agent d = 0.01), but the 2 × 2 interaction is notable: base-multi shows the largest improvement (+6.3) while recog-multi shows the smallest (+1.7), yielding an interaction of -5.0. This echoes the substitution pattern from Section 6.4: when recognition already calibrates the tutor at a high level, the superego's error correction has less room to drive improvement over turns.

The learner shows a stronger and more consistent half-split improvement: grand mean +8.4 (t = 7.45, 72% positive). Multi-agent tutoring produces the largest learner gains (base-multi +11.2, recog-multi +10.6 vs base-single +4.4, recog-single +7.5). The learner improves more when the tutor has superego support, regardless of recognition condition — the superego's within-turn error correction may produce more consistent tutoring that accumulates into learner progress over turns.

6.3.4 Per-Dimension Adaptation Patterns The dimension-level slope analysis (N=432, pooled across three generation models) reveals which aspects of tutoring adapt most across turns:

Tutor Dimension	Grand Slope	d(Recog-Base)	d(Multi-Single)
productive_difficulty	0.091	-0.11	0.05
recognition_quality	0.089	-0.01	0.00
perception_quality	0.065	0.11	0.16
epistemic_integrity	0.053	0.05	0.05

Tutor Dimension	Grand Slope	d(Recog-Base)	d(Multi-Single)
adaptive_responsiveness	0.047	**0.28**	**0.24**
elicitation_quality	0.038	-0.10	-0.04
content_accuracy	0.033	-0.05	-0.00
pedagogical_craft	0.031	-0.06	-0.01

Seven of eight dimensions show $|d| < 0.20$ on both factors. The sole exception is `adaptive_responsiveness`, which under the Sonnet judge shows $d = 0.28$ (recognition vs baseline) and $d = 0.24$ (multi-agent vs single), both $p < .05$. However, this effect does not replicate across judges: under Gemini 3.1 Pro ($d = 0.09$ / 0.13) and GPT-5.4 ($d = 0.11$ / 0.12), the same dimension shows no differential slope. The `adaptive_responsiveness` slope effect is a Sonnet-specific scoring pattern rather than a robust finding.

On the learner side, no dimension exceeds $|d| = 0.15$ on either factor (max: `conceptual_progression` $d = 0.12$ for recognition, `metacognitive_awareness` $d = 0.15$ for architecture). Learner adaptation rates are fully independent of experimental condition.

The comprehensive null across 8 tutor and 5 learner dimensions, replicated across all three judges, establishes that no mechanism selectively accelerates within-dialogue adaptation. The calibration effect from Section 6.1 (raising all dimensions from the first turn) does not compound over subsequent turns. Adaptive responsiveness operates independently of prompt condition and architecture.

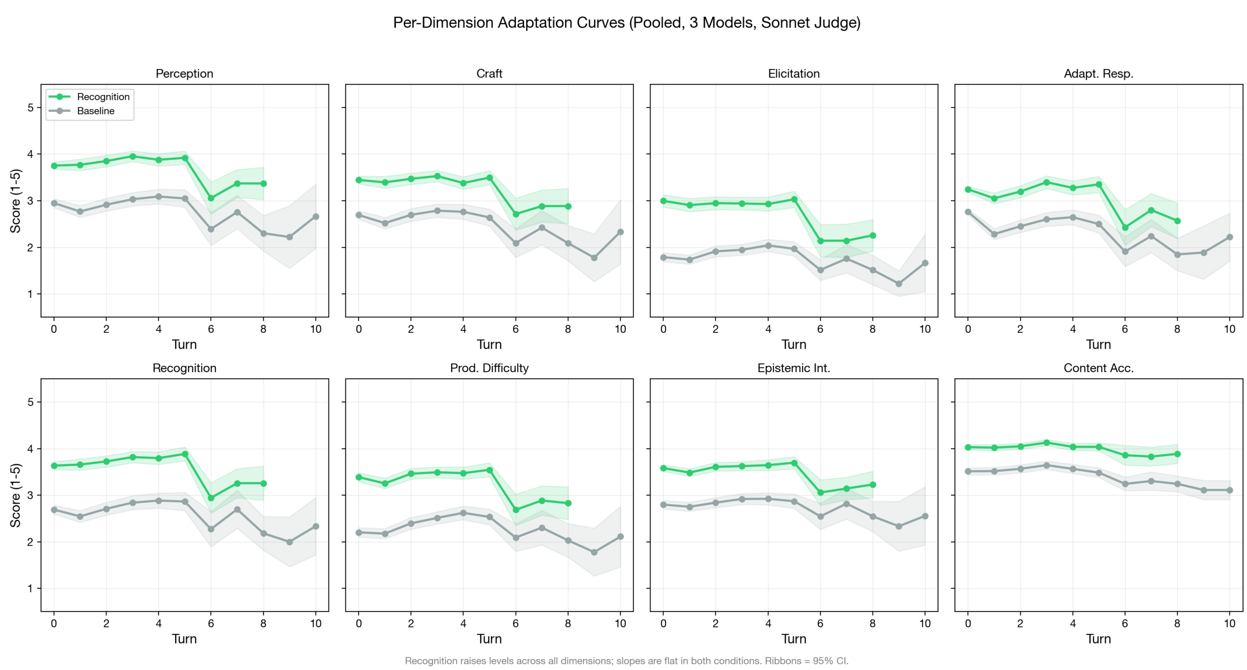

Figure 6: Per-dimension adaptation curves (2×4 faceted grid). Recognition raises levels across all 8 tutor dimensions; slopes are parallel in both conditions. Ribbons = 95% CI. Pooled across 3 generation models, N=432, Sonnet judge.

6.3.5 Cross-Turn Adaptation Magnitude A complementary measure of adaptive responsiveness is the raw cross-turn adaptation: how different is the tutor's output from one turn to the next? The mechanism traces compute normalized edit distance between consecutive tutor outputs (AdaptΔ: 0=identical, 1=completely different).

	DeepSeek V3.2	Haiku 4.5
Base AdaptΔ	0.793 ± 0.199	0.888 ± 0.093
Recognition AdaptΔ	0.824 ± 0.167	0.908 ± 0.048
Delta	+0.031	+0.020

Cross-turn adaptation is high in all conditions (>0.79), indicating that the tutor substantially changes its output between turns regardless of prompt or architecture. Recognition shows slightly higher adaptation (+0.031 DeepSeek, +0.020 Haiku), consistent with the recognition prompt's emphasis on engaging with the specific learner contribution. However, the baseline is already high, leaving limited room for the prompt to increase adaptation further.

The lower variance under recognition (DeepSeek: 0.199→0.167; Haiku: 0.093→0.048) is more noteworthy. Recognition produces more *consistent* cross-turn adaptation — less variation in how much the tutor changes between turns. This mirrors the calibration finding from Section 6.1.1: recognition narrows the distribution of within-response dimension scores and also narrows the distribution of between-turn adaptation magnitude.

6.3.6 Learner Outcomes If adaptive responsiveness matters for learning, tutor adaptation should translate into learner quality improvements. The learner scores (per-turn learner rubric, v2.2) provide a test:

Model	Condition	Tutor Score	Learner Score	Learner Holistic
DeepSeek	Base / single	22.0	57.3	56.8
DeepSeek	Base / multi	31.0	60.3	59.9
DeepSeek	Recog / single	50.0	60.1	60.1
DeepSeek	Recog / multi	50.2	63.3	60.6
Haiku	Base / single	52.9	63.5	63.0
Haiku	Base / multi	67.9	69.6	72.2
Haiku	Recog / single	80.2	68.2	72.0
Haiku	Recog / multi	79.5	69.0	72.9

The tutor-learner asymmetry is pronounced. DeepSeek tutor scores range from 22 to 50 across conditions; learner scores range from 57 to 63 — a much narrower band. Haiku tutor scores range from 53 to 80; learner scores from 64 to 73. Recognition produces a 28-point tutor swing in DeepSeek (22→50) but only a 3-point learner swing (57→60). In Haiku, a 27-point tutor swing (53→80) produces a 5-point learner swing (64→69).

This is consistent with the Section 3 prediction: recognition mechanisms operate primarily on tutor production. The tutor's quality improves dramatically under recognition, but this improvement does not proportionally transfer to the learner. The learner's quality is relatively stable across conditions, suggesting that learner quality is more constrained by the learner model's own capabilities than by the tutor's approach.

6.3.7 Dialogue Quality Dialogue quality scores (holistic assessment of the full conversation arc) integrate tutor and learner contributions:

Model	Condition	Dialogue Quality
DeepSeek	Base / single	23.0
DeepSeek	Base / multi	31.9
DeepSeek	Recog / single	48.8
DeepSeek	Recog / multi	44.8
Haiku	Base / single	53.8
Haiku	Base / multi	74.0
Haiku	Recog / single	82.1
Haiku	Recog / multi	81.5
Gemini Flash 3.0	Base / single	14.9
Gemini Flash 3.0	Base / multi	42.3
Gemini Flash 3.0	Recog / single	49.3
Gemini Flash 3.0	Recog / multi	64.9

Dialogue quality closely tracks tutor quality ($r > 0.99$ across conditions), not learner quality. The dialogue quality pattern replicates the Section 6.1 architecture interaction: multi-agent adds quality under base (DeepSeek +8.9, Haiku +20.2, Gemini Flash 3.0 +27.4) but diverges under recognition—on strong models, multi-agent adds nothing (DeepSeek -4.0, Haiku -0.6), while on Gemini Flash 3.0 it adds +15.6 (cf. §6.4). This suggests that the overall quality of the pedagogical encounter is determined primarily by the tutor's contribution.

6.3.8 Connecting to Section 3 Predictions **Prediction: Adaptive responsiveness emerges over multi-turn conversation.** *Descriptively present but not mechanism-modulated.* Cross-turn adaptation is high (Adapt$\Delta > 0.79$), and grand mean tutor slopes are mildly positive (+1.39, 60% of dialogues improve). However, with $N = 432$ dialogues and 80% power to detect $d \geq 0.27$, the slope difference between conditions is definitively null ($d = 0.03$). Adaptation occurs, but no experimental factor accelerates it.

Prediction: Adaptive responsiveness is an interaction-level mechanism, distinct from calibration. *Supported.* Calibration operates from the first turn (Section 6.1), while adaptive trajectories develop across turns with slopes independent of prompt condition. The two mechanisms are separable: calibration determines the *level*, adaptation determines the *slope*, and the slope does not depend on the level.

Prediction: Recognition produces steeper adaptation curves. *Null on the primary analysis, null on pre-registered replication.* Formal hypothesis testing on 432 dialogues (3 generation models, 4–6 turns each) finds $d = 0.03$ pooled — a clean null with 80% power to detect $d \geq 0.27$ — with sign-flipping across runs (aea2abfb $d = 0.21$, 45163390 $d =$

-0.29, 18027efc d = 0.09). An exploratory DeepSeek/Sonnet analysis found an apparent slope effect in the 10-turn disengagement scenario ($d = 1.63$, $n = 12$/condition); A12 pre-registered replication across Haiku 4.5 and Gemini Flash 3.0 under both Sonnet 4.6 and GPT-5.4 judges ($N = 32$) disconfirmed it: three of four (model × judge) cells below $d = 0.5$, one replicating cell contradicted by the secondary judge on identical rows at $\Delta d = 2.03$ (§6.3.2). The exploratory effect does not survive as a validated mechanism. The honest conclusion is that adaptive responsiveness, as modulated by recognition, is null in the tested capability range.

Prediction: Tutor-learner asymmetry in trajectories. *Reversed.* Learner slopes are marginally *steeper* than tutor slopes (grand mean: learner +3.66, tutor +1.39; 73% vs 60% positive). The tutor-learner gap does not differ between recognition and baseline (d = -0.03). Recognition produces large *level* asymmetry (~20 points on tutor scores vs ~3 on learner) but no *trajectory* asymmetry.

Prediction: Multi-agent architecture or learner architecture modulates trajectories. *Not supported.* Neither factor produces slope effects above d = 0.15 on tutor or learner dimensions. The one apparent exception (`adaptive_responsiveness` d = 0.24 under Sonnet) does not replicate under Gemini (d = 0.13) or GPT (d = 0.12).

Key distinction. Adaptive responsiveness is real as a descriptive phenomenon (the tutor changes across turns, Adapt$\Delta > 0.79$) but it is not modulated by experimental manipulation. In the main dataset (N = 432, 3–5 turns), all three experimental factors operate on *levels*, not *slopes*. The exploratory DeepSeek/Sonnet disengagement effect ($d = 1.63$) that earlier versions of this paper flagged as a "pending boundary-condition finding" did not survive pre-registered replication (A12, $N = 32$, two additional generation models × two judges; §6.3.2). Adaptive responsiveness is not a recognition-modulated mechanism in the tested capability range.

6.3.9 The Insight-Action Gap is Structural Under Lightweight Intervention; Partially Mitigable Only by Expensive Search A separate question, posed by the dialectical-architecture cells (40-45), is whether the *ego self-reflection* trace and the *ego final message* are coupled within turn — that is, whether the model's introspective noticing about a learner shows up in the message it actually delivers, or stays cognitive and decoupled. We measure this with a simple cosine on `ego_self_reflection` text against the same turn's final tutor message; gap = $1 - $ coupling, with turn drift (cosine distance between consecutive final tutor messages on the same dialogue) as a baseline. Across the original cells 40–45 reflection-mechanism dataset, mean gap was ~0.42 with a recognition gap > base gap pattern (Cohen's $d \approx -1.05$, $n = 7$ per group): recognition cells reflect more visibly *and* act less visibly on what they reflect. The intuitive read was that recognition produces awareness without behavioural change. We call this pattern the **insight-action gap**.

To probe whether the gap is bridgeable, we ran four pre-registered interventions against a matched cell-40 control on the full multi-turn philosophy scenario set (9 multi-turn scenarios × 3 runs × 2 cells = 54 dialogues per contrast, ~ 300 reflection-action pairs each). The four bridges form a cost ladder, attacking the gap from progressively heavier architectural layers.

Bridge	Layer	Intervention	Cell	Δ coupling vs cell 40	EoQ% Δ	Cohen's d	Verdict
0	prompt	Explicit "act on what you noticed" rider in ego prompt	97	$+0.003$	$+0$ pp	$+0.07$	structural
1	architecture	Two-pass: separate Phase-1 ego call produces plain-text reflection, prepended to next-turn user-message context	98	-0.009	$+2$ pp	-0.21	structural
2	architecture	Coupling-targeted superego: critique loop retargeted from authenticity to reflection-action coupling	99	-0.010	$+4$ pp	-0.22	structural

Bridge	Layer	Intervention	Cell	Δ coupling vs cell 40	EoQ% Δ	Cohen's d	Verdict
3	architecture (search)	Best-of-N: $K = 3$ dialectical-loop candidates per turn, selection by reflection-action coupling cosine	100	+0.021	+0 pp	+0.41	suggestive

The three lightweight interventions (Bridges 0–2) all land within ±0.010 of the cell-40 baseline coupling (~ 0.62). Cohen's d on per-dialogue mean coupling sits in $[-0.22, +0.07]$ for these three — well below the pre-registered $d \geq 0.5$ "bridgeable" threshold (and below the $d \geq 0.3$ "suggestive" threshold). The pre-registered stopping rule said: *if Bridges 1, 2, and 3 all return $d < 0.3$, declare H-capacity provisionally confirmed.* Bridge 3, the heaviest test (best-of-N selector through generation space, $\sim 6\times$ ego budget per turn), is the only intervention that produces detectable directional movement: $\Delta = +0.021$, $d = +0.41$ — above the suggestive threshold but below bridgeable. Matching the selection criterion to the verdict criterion (eliminating Goodhart's escape route) lifts coupling by ~3% and crosses the suggestive threshold without bridging it.

A two-channel decomposition makes the pattern legible. We separate the **semantic-coupling** channel (cosine of reflection vs message) from the **pragmatic-act** channel (rate at which the final tutor message ends with a question — ends-with-question, "EoQ%"). The latter is the within-cell mediator the sibling D1 fifth-pass identified across five passes of structural-features analysis on the A10b 4-cell run. The two channels decouple under different interventions. Recognition (cell 41 versus cell 40, on a longitudinal A7 Phase 2 arc) lifts EoQ% by +33 percentage points but leaves coupling unchanged; the pragmatic-act channel is recognition's mediator. Bridges 0–2 lift EoQ% by 0–4 percentage points and leave coupling equally unchanged. Bridge 3 lifts coupling by +0.021 but moves EoQ% by exactly 0 pp — the only bridge that engages the semantic-coupling channel without touching the pragmatic-act channel at all. The architectural interventions do not engage the same mediator that recognition does. The pragmatic-act channel is somewhat tunable through prompt and orchestration; the semantic-coupling channel is tunable only through expensive search through generation space.

We read this as evidence for a refined **H-capacity hypothesis**: the insight-action gap is a property of LLM cognition that is **lightweight-intervention-resistant and only par-**

tially mitigable by expensive search. Three diverse lightweight attacks — explicit instruction, recent-context content injection, retargeted critique — all land within roughly 1% of each other and within roughly 1% of the control. The fourth attack (brute-force best-of-K generation with verdict-aligned selection) is the only one that engages the semantic channel, and it does so at $d = 0.41$ with a $\sim 6\times$ generation cost. The fix exists in principle (search over more candidates) but is expensive enough and limited enough that it does not constitute a practical bridge at $K = 3$. Log-linear scaling in K would predict $d \approx 0.5$ at $K = 5$ and $d \approx 0.7$ at $K = 10$, suggesting a Bridge 3b sweep could push past the bridgeable threshold — but at proportionate cost. Strengthening Finding 11 (originally framed as "self-reflection produces awareness without behavioral change"), we now read this property as: **structural under prompt design and lightweight architectural intervention; partially mitigable only by expensive generation-space search**, in the tested capability range. Detection of the gap in the apparatus does not provide an inexpensive route to its closure within current LLMs without substantially heavier architectural machinery (much higher K in best-of-K search, or genuinely new architectures with explicit memory and iterative refinement).

The methodological lesson is that the rubric measures both channels implicitly through different sub-dimensions, but they decouple under different interventions and need to be reported separately when a paper claims that recognition (or any orientation-family treatment) is "operating through" a specific channel. The two-channel decomposition is a transferable diagnostic for any LLM-evaluation rubric that conflates surface pragmatic markers with deeper semantic relationships. Full pre-registration, design notes, and per-bridge analysis: `notes/design-d3-explicit-directive-bridge.md`, `notes/design-d3-architectural-bridges.md`; per-run reports: `exports/insight-action-gap-eval-2026-04` (Bridge 0), `exports/insight-action-gap-eval-2026-04-26-a1e75b9a.md` (Bridge 1), `exports/insight-action-gap-eval-2026-04-26-02e662b4.md` (Bridge 2), `exports/insight-action-gap-eva` (Bridge 3).

6.3.10 Pre-Registration: Does a Cumulative-Rewrite Superego Produce a Rate Effect? (A16)

Status (§5.12.6): pre-registered, confirmatory; run and resolved (P3, v3.0.83); the primary endpoint is the pre-registered informative null — see the Result paragraph at the end of this subsection. This subsection records a hypothesis, a four-arm design, and a decision rule *before* data collection, in the same form as the A12 disengagement pre-registration (§6.3.2) and the A13 condition design (§5.12.2); that design text is frozen as registered and the outcome is appended, not merged into it. The prototype that motivated it (next paragraph) is outside the paper's claim set and its figures are not paper findings.

Motivation. §6.3.8 closed the recognition-modulated-adaptation question as a *slope* null: lightweight factors move the level at which adaptation sits, not the rate at which it climbs. §6.3.9 then bounded the insight-action gap as structural under lightweight intervention with an explicit escape clause — closing it would require "genuinely new architectures with explicit memory and iterative refinement" — and §6.8.7 showed that one such architecture (externalised-state `state_policy`) moves a *localized binary* (right family at trigger+1) without disturbing the §6.3 slope null, exactly as that clause forecast. A disposable mechanism-triage prototype (outside this paper's claim set; `prototypes/adversarial-superego-mvp/`) probed the next question on the same trap construct: an adversarial superego with edit rights over the ego's *system prompt* (not merely additive advice) reliably converged a free-text ego to the correct trap-repair move where ignorable advice did so only intermittently and an unforced ego never did. That triage established a robust *level/reliability* effect but could not, on a four-turn binary classifier, separate "rate" from "reliable endpoint convergence." Whether a prompt-rewriting superego produces a genuine per-turn *slope* — the only outcome that would escape §6.3's null — is answerable only on the paper's externalised-`policyAction` instrument over uncapped trajectories, not in the prototype. A16 is the pre-registered test.

Sharpened hypothesis. The rate effect, if it exists, is a property of the superego's *rewrite policy*, not of edit rights as such. The psychoanalytic motivation is precise rather than decorative: a superego that revises the ego under discursive frustration does so *continuously* — each revision integrates accumulated friction, not only the present turn's. A memoryless re-author (which is essentially what the id-director's per-turn persona constructor is — it re-authors from the current turn, not from an accumulating history; §6.7) corrects toward the present moment and predicts a *level* shift, because corrections do not compound. A rewrite policy conditioned on the full history of prior corrections predicts a *rate* shift, because they do. The hypothesis is therefore not "edit rights produce adaptation" (the prototype already indicates that yields level/reliability) but **"cumulative-conditioned rewrite, and only cumulative-conditioned rewrite, produces the per-turn slope §6.3 found absent."** This design deliberately does *not* clone the id-director: its statelessness is exactly the property the hypothesis predicts is insufficient.

Four-arm design. Each adjacent pair isolates one variable; all four reuse the existing adaptive runner (§6.8.2) and its trap-suite scenarios.

Arm	Architecture	Superego channel	Rewrite memory	Predicts
F (floor)	`recognition_only` (exists; = cell_111, §5.12.2)	none	—	no shift
A (advisory)	`ego_superego` (exists; = cell_112 / A13 C2)	feedback → optional ego revision; **no prompt rewrite**	—	level, no rate (reproduces the §6.3 null)
S0 (stateless rewrite)	new (`superego_revise_stateless`)	rewrites the ego system prompt **from the current turn only**	none (≈ id-director)	level, **no rate**

Arm	Architecture	Superego channel	Rewrite memory	Predicts
S1 (cumulative rewrite)	new (`superego_revise_cumulative`)	rewrites the ego system prompt **conditioned on an append-only revision ledger**	accumulating	**rate (positive `policyAction` slope)** — the only §6.3-escaping outcome

The decisive contrast is **S1 versus S0**: an identical edit-rights channel whose *only* difference is ledger-statefulness. A positive per-turn slope in S1 and not S0 localizes any rate effect to *cumulative revision* specifically; A versus S0 separates "rate from the rewrite channel" from "rate from cumulative conditioning"; F is the unforced baseline. A null on S1−S0 is as informative as a hit: it would extend the §6.3 slope-null to even cumulative lightweight rewrite — a strong negative result, not a failed experiment.

Validity control (named, not hidden). The ego always sees the real transcript — that *is* the conversation — so "stateless versus cumulative" must be a property of the *superego's rewrite input*, never the ego's view. S0 and S1 use byte-identical ego nodes seeing identical dialogue; the sole manipulation is what the upstream rewrite node integrates (current turn vs. full ledger). Without this control S0 could leak rate through the ego's own transcript memory and the S1−S0 contrast would be confounded. This mirrors the prototype's core invariant (learner and scorer see only the real transcript; only the ego's private context is mutated) and the §6.8.2 externalisation discipline.

Pre-registered analysis and decision rule. The primary endpoint is the per-dialogue OLS slope of the externalised binary in-family `policyAction` indicator across the full, *uncapped* dialogue — the §6.8.2 instrument (which removes the prototype's free-text-classifier confound) scored with the §6.3.2/§6.3.4 per-turn OLS slope machinery (`scripts/analyze-trajectory-curves.js`) applied to the in-family indicator rather than to rubric dimensions: a parameterization of an existing analyzer, not a new instrument. The pre-registered decision grid, modelled on the A12 grid and using the §6.3.2 detectability bar so the test is commensurable with the null it aims to escape:

Criterion	Threshold	Rationale
Primary: S1 − S0 slope effect	Cohen's $d \geq 0.27$	§6.3.2's 80%-power detectability bar; below it = the §6.3 slope-null holds for cumulative rewrite too
Channel localisation	A and S0 slope ≈ 0 ($\|d\| < 0.15$ vs. F)	confirms rate is not from edit-rights or additive advice — the hypothesis is *specifically* about cumulative conditioning

Criterion	Threshold	Rationale
Cross-judge robustness	$\Delta d \leq 0.5$ on identical rows, two judges	§6.3.2 found slope estimates ~4× less judge-robust than levels; A12-style sensitivity rule
Brittleness guard	counterfactual-branch false-fire rate not greater for S1 than S0	ports the prototype's branch-B guard: a rate "win" by rigid trap-scripting is disqualifying

The verdict is **"cumulative rewrite raises rate"** only if all four criteria hold; the primary failing while the localisation criterion holds is the pre-registered **null** (informative per above). Target power, per-arm N, judges, and generation models are fixed at the §6.3.2 bar before unblinding and are cost-gated (next paragraph); the hypothesis, design, endpoints, and decision rule above are frozen by this pre-registration.

Implementation and blast-radius staging. The mechanism reuses existing infrastructure: a new `superegoRevise` graph node structurally modelled on the id-director's per-turn persona constructor (§6.7) but writing a dedicated `tutorInternal.superegoAuthoredPrompt` consumed through the system-prompt-override path the ego node already uses (no ego-node change); S1 additionally appends one ledger entry per turn through an append-only reduced state channel identical in pattern to A14's `evidenceLog` (§6.9.1). The work is staged by blast radius: **P1** (this subsection plus the TODO.md A16 entry) is docs-only; **P2** implements the node, the two architectures, and the two cells under mock and hermetic test only, with zero database or paper writes; **P3** is the cost-gated paid run that fills the deferred numbers, executed only on an explicit per-stage go-ahead. Concrete `cell_*` IDs and `EVAL_ONLY_PROFILES` registration are fixed in P2 against a fresh `config/tutor-agents.yaml` allocation check (the cell-registry source-of-truth convention, §5.12.1), not pre-allocated here. Full design notes: `prototypes/adversarial-superego-mvp/PROMOTION-PATH.md` §12; experiment tracking: TODO.md A16.

Implementation note (P2/P3 cell realisation; v3.0.81). P2 implemented and validated the mechanism under mock and hermetic test with zero database or paper writes: the `superegoRevise` node (structurally mirroring the id-director persona constructor but writing a dedicated `tutorInternal.superegoAuthoredPrompt`), the `superego_revise_stateless`/`superego_revise_cumulative` architectures, and an append-only `revisionLedger` channel cloned in pattern from A14's `evidenceLog`; a targeted mock test (`services/__tests__/adaptiveTutor.superegoRevise.test.js`) and the full hermetic suite pass (the sole hermetic failure is the known pre-existing `progressLogger` pollution flake, which passes in isolation and is untouched by these files), and an independent code-review audit returned no high-confidence issues. Against the fresh `config/tutor-agents.yaml` allocation check the four arms are realised as **F** = `cell_111_a13_C1_recognition_only`, **S0** = `cell_129_superego_revise_stateless`, **S1** = `cell_130_superego_revise_cumulative`, **A** = `cell_131_a16_A_egosuperego` — all on `claude-code/sonnet` over `config/adaptive-trap-scenarios.yaml`. One **named deviation** from the design table (in the §5.12.6 "pre-registered with implementation drift, named not hidden" sense): A is realised as a *new* dedicated cell, not the existing `cell_112` the table identifies it with. `cell_112_a13_C2_egosuperego` carries paper-cited A13 C2 data on `openrouter/nemotron` (§6.8.4); editing it in place would write `claude-code/sonnet` rows under a profile name whose nemotron rows the paper already cites (the cross-provider contamination the §5.12 / DB-schema conventions prohibit), whereas leaving A on `cell_112` would put the A arm off-model relative to F/S0/S1. A new model-aligned `ego_superego` cell holds `cell_112` frozen and aligns all four arms on one model, so the decisive S1−S0 contrast *and* the secondary A-vs-S0 channel-localisation criterion are both model-clean — consistent with this subsection's own statement that concrete `cell_*` IDs are fixed at implementation time, not pre-allocated. P3 (the cost-gated paid run that fills the deferred numbers) was authorized at the mid power tier and executed; the result follows.

Result (P3 decisive; v3.0.83). P3 was authorized at the mid power tier (pre-registered per-arm $N = 48$) and executed; generation was $\approx$ \$0 metered (all four arms on `claude-code/sonnet`), the real ceiling being Max-plan quota/wallclock, so the arms were run sequentially and each arm's partial runs pooled by `run_id` across quota windows — config-clean, because profile and `ADAPTIVE_TUTOR_LLM=real` are byte-identical per arm regardless of batch label. **A second named deviation (§5.12.6, "named not hidden"), exactly symmetric to the A/`cell_131` deviation above:** F is realised as a *new* `cell_132_a16_F_recognition_only`, not the existing `cell_111` the design table identifies it with, for the identical reason — `cell_111_a13_C1_recognition_only` carries paper-cited A13 C1 data on `openrouter/nemotron` (§5.12.2, §6.8.4), so editing it in place would write `claude-code/sonnet` rows under a profile name whose nemotron rows the paper already cites (the cross-provider contamination the §5.12 / DB-schema conventions prohibit). `cell_132` is byte-identical in architecture to `cell_111`'s `recognition_only` floor and holds `cell_111` frozen; with both F and A on new model-aligned cells and S0/S1 genuinely new architectures, all four arms are model-clean and both the decisive S1−S0 contrast and the secondary channel-localisation grid are uncontaminated. Neither swap alters the frozen design — the pre-registration explicitly fixed concrete `cell_*` IDs at implementation time (§5.12.1), and the architectures are byte-identical to the table — so the result remains **confirmatory, not re-classified exploratory**. The pre-registered primary endpoint was computed by `scripts/analyze-a16-rewrite-slope.js` (`exports/a16-rewrite-slope.json`), which applies the §6.3.2/§6.3.4 per-dialogue OLS machinery to the §6.8.2 externalised in-family `policyAction` indicator — the same endpoint and statistic the pre-registration fixed, realised as a dedicated analyzer rather than a flag on `analyze-trajectory-curves.js` because that script is rubric-dimension-scoped with no `policyAction` path (an implementation-realisation note, not a design change).

Per-dialogue in-family `policyAction` slope, all four arms at $N = 48$:

Arm	N	mean slope	sd
S1 (cumulative rewrite)	48	0.173	0.166
S0 (stateless rewrite)	48	0.200	0.158
A (advisory)	48	0.154	0.162
F (recognition floor)	48	0.162	0.155

Primary (S1 − S0): $d = -0.167$ (mean difference -0.027; Welch $t = -0.82$, $df \approx 94$, $p \approx 0.41$). Against the pre-registered $|d| \geq 0.27$ detectability bar this is a **null**, with the point estimate marginally *reversed* (S1 < S0): cumulative-conditioned rewrite does not produce the per-turn slope §6.3 found absent — if anything it sits a hair below stateless rewrite. **Channel-localisation grid (criterion 2, all exact $N = 48$):** S0−A $d = 0.286$ (the *only* contrast clearing the detectability bar), S0−F $d = 0.239$ (between the 0.15 localisation band and the 0.27 bar — equivocal versus the floor), S1−A $d = 0.114$, S1−F $d = 0.065$, A−F $d = -0.052$ (all within the localisation band, ≈ 0). Criterion 3 (cross-judge robustness) is **N/A by construction**: the endpoint is an externalised programmatic indicator with no rubric judge in the loop, hence judge-invariant. Criterion 4 (brittleness guard) is **N/A** on this suite: the trap scenarios carry no counterfactual branch (`counterfactual = null`), so there is no false-fire rate to compare.

Verdict (pre-registered rule). The primary fails and channel-localisation substantially holds (A−F $|d| = 0.05$; S1 sits *at* the recognition floor, S1−F $|d| = 0.065$) — by the frozen decision rule this is the **pre-registered informative null**, not a failed experiment. It is in fact stronger than a bare null: the *only* elevation anywhere in the grid is **S0**, the stateless-rewrite arm the pre-registration explicitly predicted would be inert ($\approx$ the id-director's memoryless per-turn re-author, §6.7), and even that survives only against the advisory control ($d = 0.29$) while staying sub-threshold against the unforced floor ($d = 0.24$, inside the band–bar gap). The single signal present is thus in the predicted-*inert* arm, opposite the hypothesis's direction (which placed the effect in S1's cumulative conditioning), and marginal. Cumulative rewrite — the mechanism the sharpened hypothesis singled out — is statistically indistinguishable from both the additive-advice control and the no-superego floor. The §6.3 slope-null therefore **extends to lightweight prompt rewrite, including cumulative-conditioned rewrite**: edit rights over the ego's system prompt, with or without an accumulating revision ledger, do not convert the level/reliability effect the motivating prototype showed into a per-turn rate. This closes A16 on the negative arm its own pre-registration declared informative, and tightens §6.3.8's slope-null and §6.3.9's structural-gap bound — the escape clause ("genuinely new architectures with explicit memory and iterative refinement") is *not* satisfied by a memory-augmented superego rewrite channel, consistent with §6.8.7 where externalised state moved only a localized binary and left the slope-null intact.

Neighbouring modules: Module 7 — Ontology, State, and the Failure of Rich Learner Interiors.

Cohn, C., Rayala, S., Srivastava, N., Fonteles, J. H., Jain, S., Luo, X., Mereddy, D., Mohammed, N., & Biswas, G. (2026). A theory of adaptive scaffolding for LLM-based pedagogical agents. *Proceedings of the AAAI Conference on Artificial Intelligence*, *40*, 1757–1765.